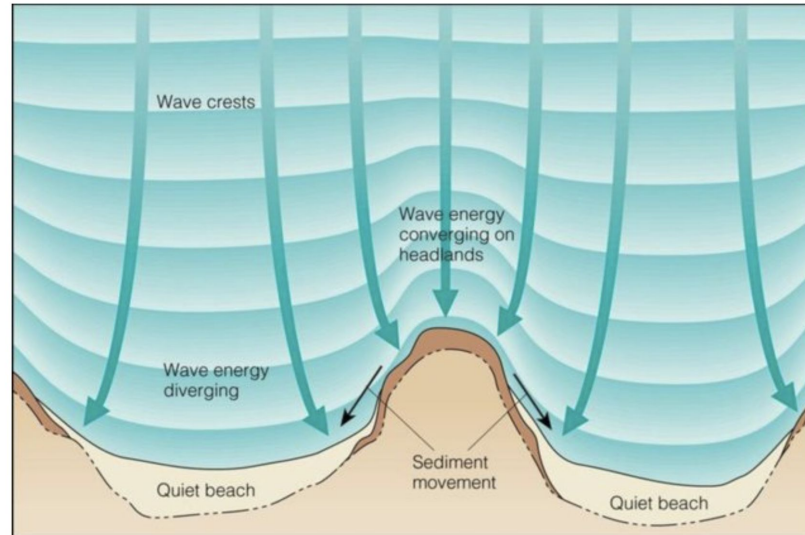


GP 245 Project

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Scientific background and motivation

- How well do finite-difference shallow water models reproduce theoretical tsunami propagation?
- Can the linear 2D shallow water equations capture wave focusing along certain segments of the coastline when using real bathymetry, specifically at Mavericks, CA (famous for large waves)?



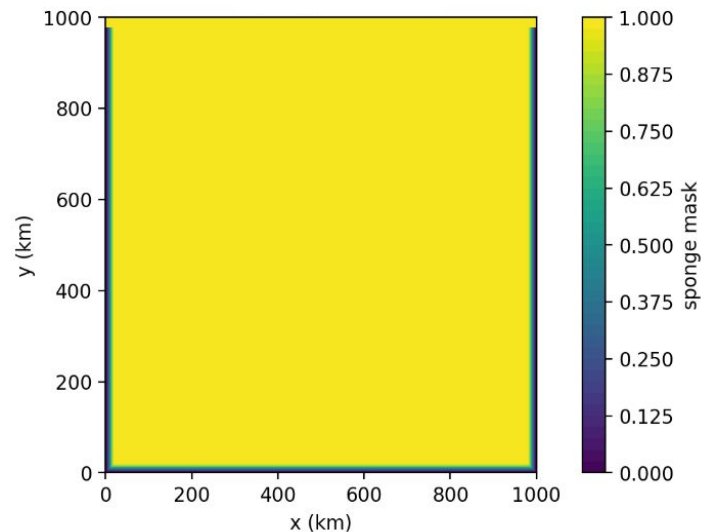
Garrison 2005

Equations and boundary conditions

mass conservation:
$$\frac{\partial \eta}{\partial t} = -\frac{\partial(Hu)}{\partial x} - \frac{\partial(Hv)}{\partial y}$$

momentum conservation:
$$\frac{\partial u}{\partial t} = -g \frac{\partial \eta}{\partial x}$$
$$\frac{\partial v}{\partial t} = -g \frac{\partial \eta}{\partial y}$$

Symbol	Description
$\eta(x, y, t)$	Free surface elevation (m)
$u(x, y, t)$	Velocity in x -direction (m/s)
$v(x, y, t)$	Velocity in y -direction (m/s)
$H(x, y)$	Water depth (m)
g	Gravitational acceleration
x, y	Spatial coordinates (m)
t	Time (s)



Using cosine taper (from 1 to 0) to apply damping at ocean boundaries for 20 grid cells from the domain boundaries

Numerical method (MacCormack method) and computational implementation

Predictor step (forward differences on current state):

$$u_{i,j}^* = u_{i,j}^n - \frac{g\Delta t}{\Delta x}(\eta_{i+1,j}^n - \eta_{i,j}^n)$$
$$\eta_{i,j}^* = \eta_{i,j}^n - \Delta t \left[\frac{H_{i+1,j}u_{i+1,j}^n - H_{i,j}u_{i,j}^n}{\Delta x} + \frac{H_{i,j+1}v_{i,j+1}^n - H_{i,j}v_{i,j}^n}{\Delta y} \right]$$

Corrector step (backward differences on predicted state):

$$u_{i,j}^{n+1} = \frac{1}{2} \left(u_{i,j}^n + u_{i,j}^* - \frac{g\Delta t}{\Delta x}(\eta_{i,j}^* - \eta_{i-1,j}^*) \right)$$
$$\eta_{i,j}^{n+1} = \frac{1}{2} \left(\eta_{i,j}^n + \eta_{i,j}^* - \Delta t \left[\frac{H_{i,j}u_{i,j}^* - H_{i-1,j}u_{i-1,j}^*}{\Delta x} + \frac{H_{i,j}v_{i,j}^* - H_{i,j-1}v_{i,j-1}^*}{\Delta y} \right] \right)$$

CFL condition for timestep size:

Using 0.2 for simple cases
and 0.1 for more complex
Mavericks simulation
(smaller timestep)

- Parallelization using Numba: each column of grid is assigned to a core

Possible simulation setups

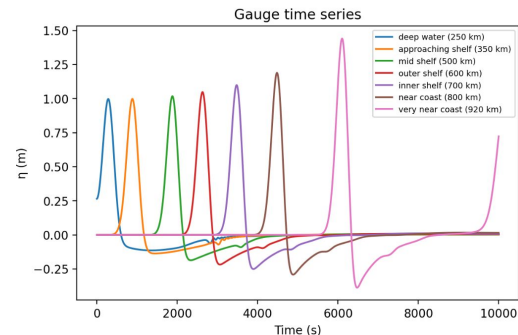
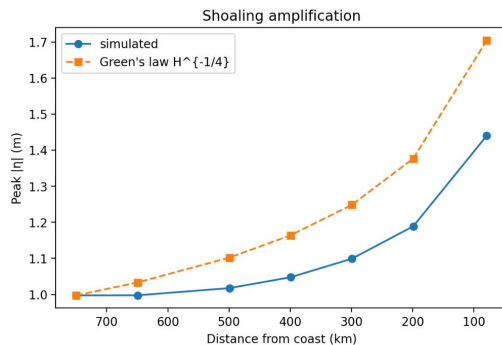
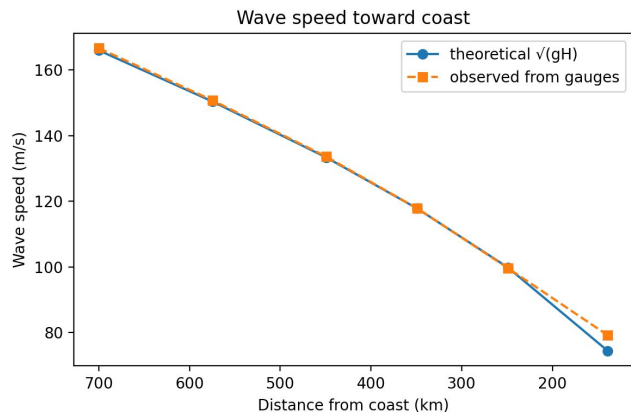
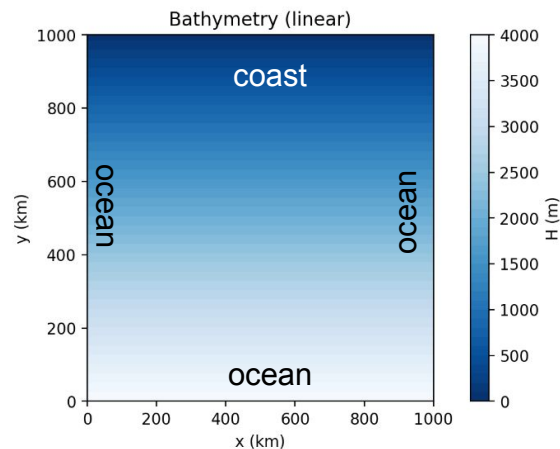
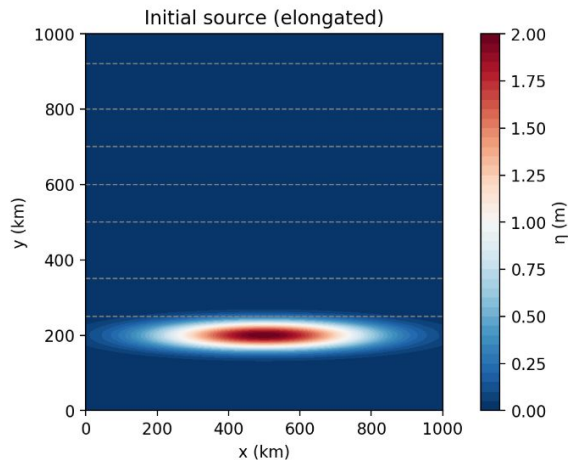
Source options: 1) Circular, 2) elongated

Seafloor options: 1) constant slope, 2) linear slope, 3) real bathymetry

Simple case example

Source: elongated

Seafloor: linear slope



Compare to theory:

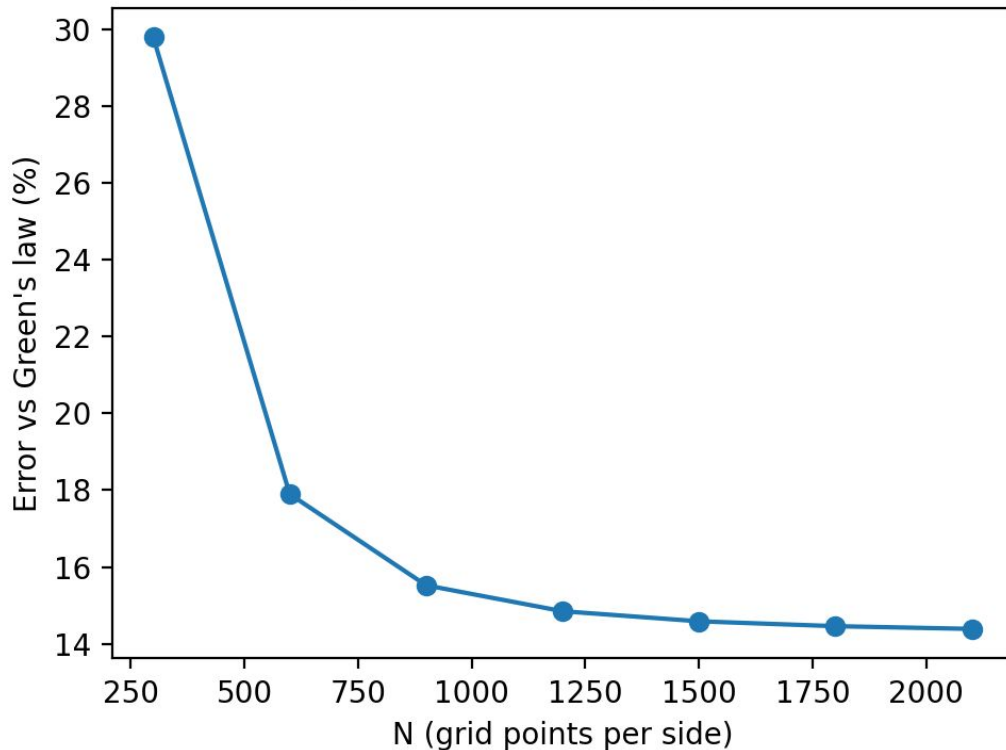
Wave speed:

$$c = \sqrt{gH}$$

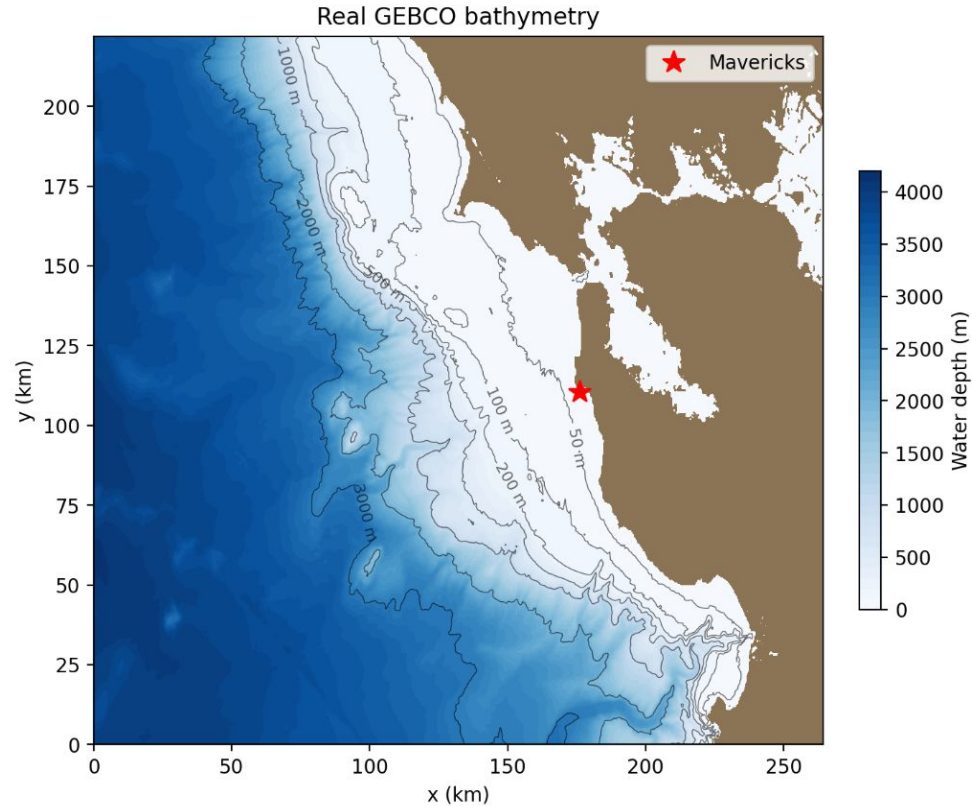
Green's law:

$$\frac{\eta_2}{\eta_1} = \left(\frac{H_1}{H_2} \right)^{1/4}$$

Grid convergence for simple case of elongated source and linear bathymetry

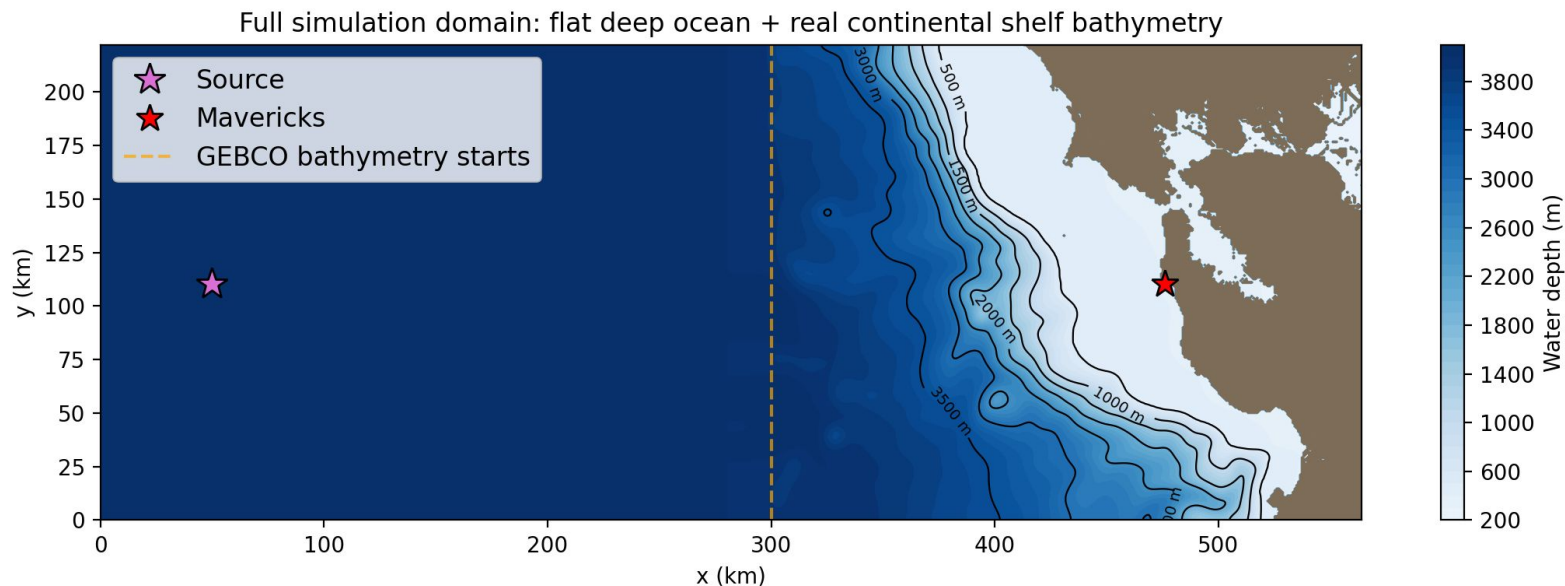


Does the bathymetry near Mavericks focus tsunami wave energy onto that stretch of coastline compared to adjacent areas?



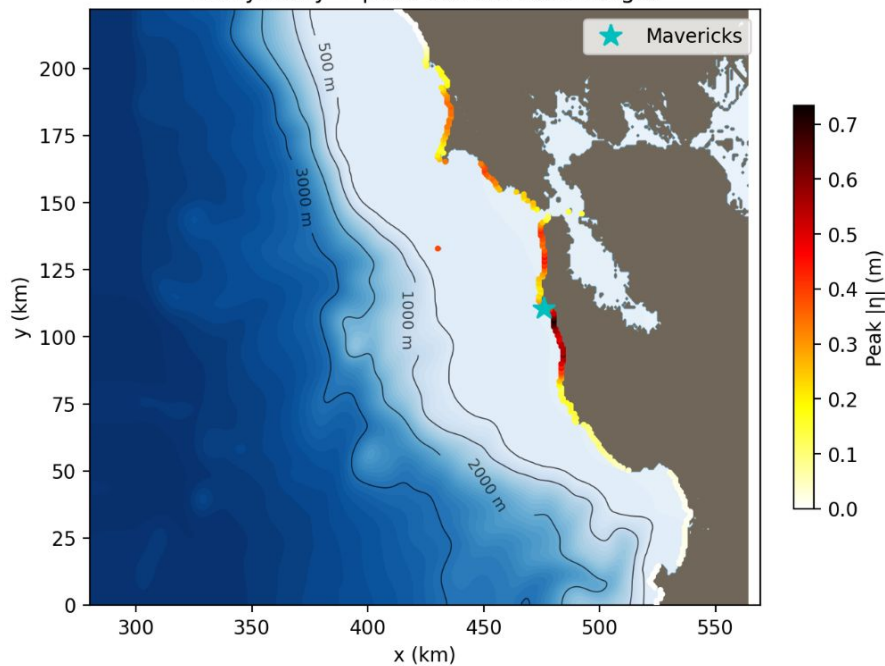
Bathymetry for simulation

- Nearshore region was replaced by linear slope from 500 to 200m depth to handle stability issues in shallow water
- Circular source

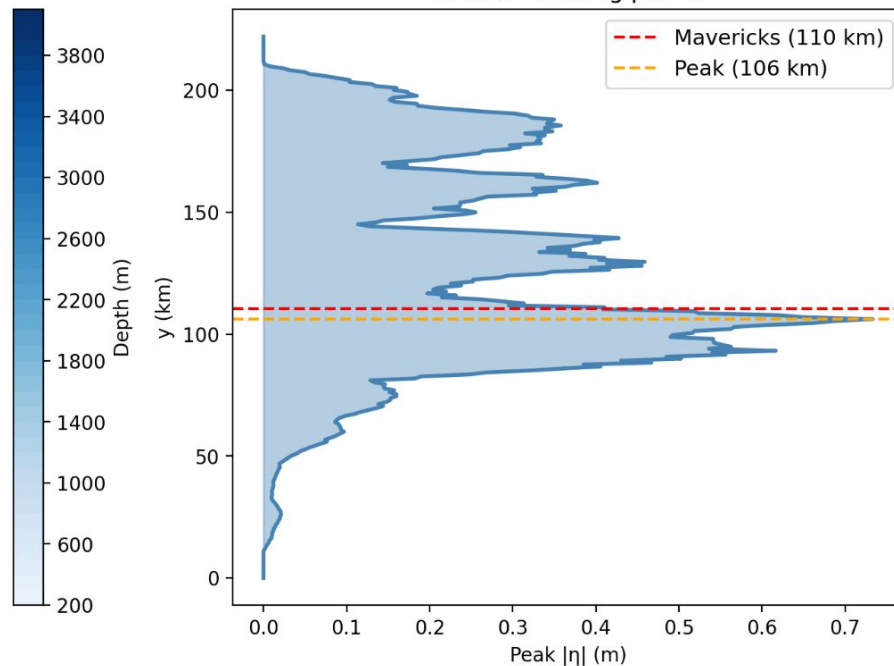


Max simulated wave amplitude is near Mavericks

Bathymetry + peak coastal wave height



Coastal focusing profile



Lessons learned and potential future improvements

Lessons learned:

- Issues with sharp gradients at water-land interface, solver handles gradual changes better, had to replace bathymetry with linear ramp
- Green's law: expected my simulation to match theoretical prediction better, error is likely due to geometric spreading in 2D. Green's law may not be the appropriate solution to compare to

Potential improvements:

- Use nonlinear shallow water equations to capture runup and wave breaking in the nearshore
- Use smaller CFL condition or a different, more robust numerical solver to resolve shallow coast